

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (CE)

Nov 2012

CE 207 Java Programming

Date: 05.11.2012, Monday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) What is a keyword? How does a keyword differ from an identifier? [04]
(b) Explain static blocks in java. [03]
- Q - 2 (a) What is automatic type promotion in expression? [03]
(b) Explain how static keyword modifies the role of method and variable with example. [04]
(c) Write a program that specifies whether a given number (x) falls in one of the [07]
following categories (give x a value from the code, don't read from user):
0 to 9, 10 to 19, 20 to 29, None of the categories

OR

- Q - 2 (a) What is data type? List the data types available in java. [03]
(b) Explain toString() method of Object class. [04]
(c) Write a program that asks the user to enter two numerical values (integers) and then [07]
select an operation (addition, subtraction, multiplication and division) then prints the
result based on operation selected.
- Q - 3 (a) Explain the Bitwise Operators. [03]
(b) Write the difference between method overloading and constructor overloading. What [04]
is method overriding?
(c) Write a program to create interface A. In this interface declare two method meth1 and [07]
meth2. Implements this interface in another class named MyClass. Can interface have
an instance variable?

OR

- Q - 3 (a) What is ?: (ternary) Operator? How is it similar with if else statement? [03]
(b) What is data hiding? How data can be hidden within class? [04]
(c) What is an abstract class? Why object can't be created of abstract class? Can static [07]
method be an abstract?

SECTION – II

- Q - 4 (a) Explain the life cycle of Thread with neat diagram. [05]
(b) Explain *break* and *continue* keywords. [02]
- Q - 5 (a) List the differences between applet and AWT. Also write the differences between Component and Container classes. [07]
(b) Write a program for creation of user defined exception. [07]

OR

- (b) Write a program to find the given string is palindrome or not. String will be inputted at run time by user. [07]
- Q - 6 (a) "String class is immutable java class." Justify the statement. Explain the following methods of String class: [07]
`charAt( )`, `getChars( )`, `getBytes( )`, `toCharArray( )`

OR

- (a) Explain the following methods related to File class: [07]
`getName( )`, `getPath( )`, `getAbsolutePath( )`, `getParent( )`, `exists( )`, `canWrite( )`, `isDirectory( )`
- Q-6 (b) Explain event delegation model of java. [07]
